

Shaden Shanab

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Senior full-stack data scientist with a proven record of building and shipping production-grade AI systems across retail, fintech, SaaS, and data governance domains. Certified by TensorFlow and recognized by Jordan's Ministry of Higher Education as the top AI student in 2023, with expertise spanning end-to-end ML pipelines, forecasting, LLM engineering, agentic AI, and conversational systems.



Experience

April 2026 – Present

AI Consultant | Freelance

- Partnered with businesses across retail, SaaS, marketing, and customer experience to identify pain points, design AI solutions, and deliver end-to-end implementations covering architecture, LLM integration, and deployment.

Ai-Assisted Data Platform

- Ingests messy datasets and profiles them for completeness, type inference, distributions, duplicates, and structural issues.
- Differs datasets across uploads and flags statistically significant shifts and anomalies.
- Runs automated per-dataset and cross-dataset analysis, generating dashboards with plots and evidence for each finding.
- Self-serve chart and dashboard builder, with metric-level comparison across datasets.
- LLM agent with schema-grounded function calling that translates natural-language questions (English or Arabic) into SQL and returns results with analysis, charts, and the query used.
- Governance layer storing definitions, ownership, bilingual metadata, and lineage attached to the schema.

Voice-To-Voice Customer Service Agent

- Agentic voice-to-voice pipeline combining real-time STT, dialect-aware TTS, and an LLM orchestration layer with function calling for live actions like database lookups, API calls, and call transfer.
- Arabic dialect classification and dialect-matched response generation, VAD, barge-in support, low-latency turn-taking.
- Sentiment detection integrated into the conversation loop to flag frustration and trigger escalation.

Contact Center Management Platform

- Arabic-language portal unifying conversation management, claims, complaints, renewals, and sales performance for agents and managers.
- Live conversation queue and kanban-style claims pipeline with ai and human lanes, sentiment-based prioritization, and full case context for handoffs.
- Analytics suite tracking claim volume, sentiment, satisfaction, payments, renewals, and sales conversion.
- Ai-to-human escalation logic based on customer sentiment, explicit request, or ai confidence limits.

Call Analytics Platform

- Contributed to an ai-powered call analytics platform that detects emotional tone from voice, scores service quality, and generates instant, detailed reports in arabic.

December 2025 – April 2026

Senior Data Scientist | Governata

- Drove personal data detection accuracy from 26% to 92% by engineering end-to-end PII detection pipelines using NLP and ML techniques across large-scale enterprise datasets.
- Engineered automated pipelines for FK relationship mapping and data classification across large-scale enterprise datasets.
- Executed the full data governance lifecycle including data exploration, profiling, classification, lineage tracking, and policy enforcement.
- Collaborated with legal, IT, and business stakeholders to translate regulatory requirements into auditable technical systems and documented governance standards.

January 2025 – December 2025

Data Scientist | REVEST

- Drove end-to-end development of innovative AI solutions, from ideation and business analysis to prototyping, deployment, and impact measurement.
- Designed and implemented advanced ML, deep learning, and generative AI applications, including time series models for customer behavior and sales forecasting, personal analyst chatbots, and LLM evaluation frameworks.
- Created intelligent systems that analyze business inefficiencies, recommend actionable strategies, and evaluate ROI and long-term business impact.
- Led and mentored the team on high-impact projects while independently driving project foundations, strategy, and execution.

September 2023 – December 2024

Data and Analytics Specialist | INGOT BROKERS

- Built and maintained the core Python codebase for ETL processes and third-party integrations (APIs, cron jobs, data transformations), ensuring scalability and maintainability across multiple data sources.
- Implemented ETL pipelines and data warehousing solutions that improved operational efficiency and data integration.
- Applied SQL transformations, dimensional modeling, and data preprocessing to automate reporting and dashboards.
- Led a transformation of the data syncing technology, increasing speed by 70% and improving system stability.

 Education

MAY 2024

TensorFlow Developer Certificate | TensorFlow

Received for demonstrating proficiency in building and training neural network models using TensorFlow, encompassing diverse domains including data science, image processing, time series analysis, and natural language processing (NLP).

MARCH 2021 – JULY 2023

Artificial Intelligence | Abdul Aziz Al Ghurair School of Advanced Computing (ASAC)

Ranked 1st.

GPA: 95.3% (Distinction)

BTEC Level 5 Associate’s Degree. Covered topics such as data analytics, database programming, data mining, machine learning, applied analytical models, object oriented programming, data structures & algorithms, with hands-on projects in them all.

JULY 2022 – JANUARY 2023

Software Development in Python Bootcamp | Code Fellows

Completed an intensive software development bootcamp with a focus on Python, gaining valuable experience in full-stack development, database programming, and software engineering best practices.

 Skills

- Python, Java, JavaScript, Arabic (Native), English (Fluent).
 - Voice/Speech AI: Speech-to-Text, Text-to-Speech, Voice Activity Detection.
 - LLM Orchestration: Agentic workflows, Function Calling, RAG.
 - NLP: Natural Language Processing, Named Entity Recognition, PII Detection.
 - Machine learning and deep learning: TensorFlow, PyTorch, Scikit-learn, Ollama, HuggingFace, Pydantic-AI, Prophet, OpenAI, Mistral, Gemini.
- Web Development: Django, Flask, Fast API, Node.js, React, REST API, Next.js, TailwindCSS, HTML, CSS.
 - Database: MySQL, PostgreSQL, SQLite, MongoDB, Clickhouse, DBT, Redis.
 - Data Engineering: ETL processes, Data Modeling, Data Governance, Data Profiling and Data Warehousing.
 - Big Data & Cloud Services: AWS, Kafka, Redshift.
 - Data analytics and data visualization: Excel, PowerBI, pandas, matplotlib, plotly, and NumPy